Raksha Narasegowda

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EDUCATION

Master of Science, Computer Science and Engineering | Santa Clara University | GPA: 3.6/4.0 — **Jan 2023 - Dec 2024**

Relevant Coursework: Design and Analysis of Algorithms, Machine Learning, Operating Systems, Computer Architecture, Database Systems

Bachelor of Technology, Computer Science and Engineering | REVA University | GPA: 4.0/4.0 — **Aug 2018 - July 2022**

Relevant Coursework: Cloud Computing, Data Structures, Data Science, Distributed Systems, Object Oriented Programming, NLP

TECHNICAL SKILLS

- **Programming Languages**: Python, Java, C, C++, JavaScript, R, Go, Kotlin
- **Web Development : Frontend -** Angularjs, React, Flutter, HTML, CSS | **Backend** - Python (Fast API, Flask), Django, Node/Node.js (Express)
- **Databases :** MongoDB, Postgres, DynamoDB, NoSQL, SQL, Kafka, Spark, Apache Hadoop, Redis
- **Transport Layers :** Graphql, Restful/Rest API, Web Sockets, TCP/IP, UDP, ARP, DNS, DHCP
- **Cloud Platform & DevOps Tools:** Docker, Kubernetes, AWS(EC2, EKS, S3, ECS, IAM), GCP, Jenkins, Agile, Scrum, CI/CD, Terraform, Git
- **Data Science :** TensorFlow, PyTorch, Keras, Pandas, numpy, SciPy, scikit-learn, NLTK, PySpark | **OS -** Unix, Windows

EXPERIENCE

Software Developer | *Frugal Innovation Hub* | *Python, PostgreSQL, AWS, Docker, Plotly, jenkins* — **Mar 2023 - Present**

- Engineered a scalable and distributed data processing pipeline using **Python, GeoPandas,** and **PostGIS** to handle 20+ GB of satellite geospatial data, improving system throughput and boosting the accuracy of climate risk predictions by **60%**.
- Optimized microservices architecture by containerizing **AWS Lambda** functions and applying vectorized geospatial transformations, reducing API response latency from minutes to seconds and enhancing system scalability by **70%**.
- Built and deployed end-to-end ML workflows with Docker and **Jenkins** CI/CD pipelines, ensuring repeatable model deployment and improving development velocity and production reliability across cloud environments.
- Improved system design and backend usability for analytical tools by integrating Python-based plugins with QGIS and performing design/code reviews, unit testing, and error profiling— reducing data interpretation and decision-making time by **30%** for end-users.

Associate Software Engineer | *DXC Technology* **|** *Java, MongoDB, Django, Terraform, Google Cloud Platform* — **June 2022 - Nov 2022**

- Reduced customer onboarding time by **40%,** by designing and developing scalable backend microservices and **RESTful APIs** for CRM workflows using **Java** and **Django** respectively, along with **MongoDB** and **Terraform** on **GCP**, applying distributed system design principles.
- Improved system uptime by **80.9%** by leading Jenkins-based CI/CD pipeline integration, enforcing software craftsmanship via automated testing and performance profiling under real-time API loads.
- Streamlined data integration across 3+ systems by developing **Django-based RESTful APIs** and collaborating closely with frontend teams using ReactJS in Agile sprints, boosting user engagement by **30%**.
- Reduced regression bugs by **25%** by instituting design reviews, unit/integration testing, and Git version control best practices, ensuring modular, maintainable code delivery leveraging Django’s Model-View-Template (MVT) architecture.

PROJECTS

Personal Finance Tracker | *React.js, HTML, CSS, Node.js, Express.js, MongoDB*

- Built a full-stack personal finance web app using **React.js, Node.js, Express.js,** and **MongoDB**, enabling users to track income, expenses, and budgets in real time, and improving personal budgeting efficiency.
- Implemented secure **JWT-based** authentication and role-based access control, ensuring safe multi-user support and protecting sensitive financial data from unauthorized access.

Automated Backup System | *Python, Golang, Google Cloud Storage, Unix, Docker*

- Improved system reliability and performance, by designing and deploying a distributed backup system using **Python** and **Go,** leveraging Go’s concurrency to accelerate data transfer and optimize resource utilization.
- Enabled fault-tolerant and reproducible backups, by implementing the Paxos consensus algorithm for distributed **cron scheduling** across Unix systems, and containerizing the workflow with **Docker** to ensure consistent deployments and redundant storage in **Google Cloud**.

Shrinkify – *Shrink it. Share it. Scale it* | *RESTful API, Express.js, MongoDB, Kubernetes*

- Built a scalable and high-availability URL shortening service, by developing a **RESTful API** with **Express.js** and storing efficient short-to-long URL mappings with **MongoDB**, enabling real-time redirection and analytics tracking.
- Ensured system reliability and traffic resilience, by integrating **Kubernetes** for container orchestration, allowing the service to handle high traffic loads with minimal downtime.

CERTIFICATIONS & COURSES

Google: Google Data Analytics | **AWS:** AWS Certified Cloud Practitioner | **Microsoft:** Azure Fundamentals